

8

Causal Models – Assumptions and Challenges

Welcome to *Chapter 8*.

In *Chapter 7*, we demonstrated how to leverage the power of the DoWhy library to estimate causal effects. The goal of this chapter is to deepen our understanding of when and how to use causal inference methods.

We'll review some of the assumptions that we introduced earlier in *Chapter 5* and we'll discuss some more assumptions in order to get a clearer picture of the challenges and limitations that we might face when working with causal models.

By the end of this chapter, you will have a good understanding of the challenges that you may face when implementing causal models in real life and possible solutions to these challenges.

In this chapter, we will cover the following:

- The challenges of causal inference methods
- Identifiability
- Positivity assumption
- Exchangeability/ignorability assumption
- Modularity (independent mechanisms)
- Consistency
- SUTVA
- Selection bias

I am the king of the world! But am I?

So far, we've seen what we can achieve by using causal models on observational data and we promised that we'll see them do even more impressive things. All this might feel pretty powerful!

In this section, we're going to discuss important challenges that we might face when using causal inference methods in practice.

We'll start by sketching a broader context. After that, we'll explicitly define the concept of **identifiability**. Finally, we'll discuss some popular challenges faced by practitioners:

- A lack of a priori knowledge of causal graphs
- Insufficient sample sizes
- Difficulties with verifying the assumptions

In between

Ancient Greek mythology provides us with a story of Ikaros (also known as Icarus) and his father, the Athenian inventor Daidalos (also known as Daedalus). Daidalos wants to escape from Crete – a Greek island where he's trapped. He builds two sets of wings using feathers and wax. He successfully tests his wings first before passing the other set to his son.

Before they fly, Daidalos advises his son not to fly too low, nor too high: “*for the fogs about the earth may weigh you down and the blaze from the sun are going to melt your feathers apart*” (Graves, 1955). Ikaros does not listen. He's excited by his freedom. Consumed by ecstatic feelings, he ascends toward the sun. The wax in his wings starts melting and – tragically – he falls into the sea and drowns.

The history of causality starts in a place of big dreams and seemingly impossible goals. From broad Aristotelian concepts of four causes, through the ages of religious and philosophical reflection, through David Hume's ideas, up to modern econometrics and Pearl's *do*-calculus, we've come a long way, and, just like Daidalos, we learned to fly.

It's likely that we haven't yet reached the *Causal Promised Land* and that we will continue to learn and grow in our causal journey in the coming years.

In the meantime, we have an amazing set of causal tools in our hands. In order to use them effectively, we need the awareness of their strengths and their limitations. The path is somewhere *in between*. By knowing the assumptions behind causal methods and the challenges that we might face on our way, we can improve our businesses and our communities, bringing more growth, more knowledge, and more insights.

Understanding the assumptions that we discuss in this chapter will help you achieve two things:

- Become a better (data) scientist (even if your primary goal is not to conduct causal analyses)
- Learn how to effectively use causality without over- or undervaluing its potential

Ready?

Let's go!

Identifiability

One of the core concepts that we'll use in this chapter is **identifiability**. The good news is that you already know what identifiability is.

We say that a causal effect (or any other causal quantity) is **identifiable** when it can be computed unambiguously from a set of (passive) observations summarized by a distribution $P(V)$ and a causal graph G (Pearl, 2009).

In other words, if we have (1) enough information to control for *non-causal* information flow in the graph and (2) enough data to estimate the effect of interest, the effect is identifiable.

Let's unpack that.

Do you remember *Chapter 6* and our conversation on *d*-separation and *do*-calculus?

The first condition is achievable by blocking all the paths that are *leaking* non-causal information using the rules of *do*-calculus and the logic of *d*-separation. This is often possible using the back-door criterion, the front-door criterion, instrumental variables, or the general rules of *do*-calculus.

That said, sometimes it is just impossible.

The important thing is *not* to pretend that it's possible when we know it's not. In such a case, using causal methods can bring more harm than good. One of the main challenges is that sometimes we simply don't know. We'll learn more about this in the last part of this section.

The second condition has two faces.

First, any estimator needs to have a large enough sample size to return meaningful estimates. Second, we need to make sure that the probability of every possible value of treatment in our dataset (possibly conditioned on all important covariates) is greater than 0. This is known as the **positivity assumption** and we'll discuss it in the next section.

Now, let's talk about a couple of challenges causal data scientists face.

Lack of causal graphs

In a recent survey that I conducted on LinkedIn, I asked the participants to imagine that they could ask a world-class causality expert one question. 28% chose to ask a question related to obtaining causal graphs in real-world scenarios.

The lack of a causal model can be a major challenge in implementing causal inference techniques in practice.

There are three things that people usually do to obtain a causal graph for a problem with an unknown causal structure:

- Use domain expertise
- Use causal discovery techniques
- Use a combination of both of the above

High quality trustworthy **domain expertise** is often the least risky option yet might be the hardest to obtain. **Causal discovery methods** are usually cheaper, but verifying their results can be challenging.

In many cases, a combination of domain knowledge and structure learning algorithms might be the best option. Some causal discovery methods allow us to easily incorporate our domain knowledge and then they perform the search over the remaining parts of graph search space for us (more on this topic in *Part 3, Causal Discovery*). This can lead to truly amazing results, yet there are no guarantees.

If your data comes from a **natural experiment**, you can also try some of the methods from the field of econometrics, such as **synthetic controls**, **regression discontinuity**, or **difference-in-differences**. We'll briefly discuss one of them (with a Bayesian flavor) in *Chapter 11*.

Not enough data

An insufficiently large sample size is not a uniquely causal problem. Any statistical parameter estimate becomes biased under an insufficiently large sample size.

Depending on the method we choose, some causal methods might require more data than others. If we use the **double machine learning (DML)** technique with neural network estimators, we need to be prepared for neural-network-style sample sizes.

Nonetheless, sample size can also be an issue in the case of very simple models. I conducted a simple experiment so we could see it in action. The full code for this experiment is in the notebook for *Chapter 8* (<https://bit.ly/causal-ntbk-08>). Here we only present the methodology and the results.

Let's take the graphical model from the previous chapter (check *Figure 8.1* for a refresher).

Understanding the sample size – methodology

Let's take four different sample sizes (30, 100, 1,000, and 10,000), and for each sample size, generate 20 datasets from the model presented in *Figure 8.1*. We fit two models for each dataset: one based on simple linear regression and one based on a DML estimator, powered by two gradient-boosting models and a cross-validated lasso regression:

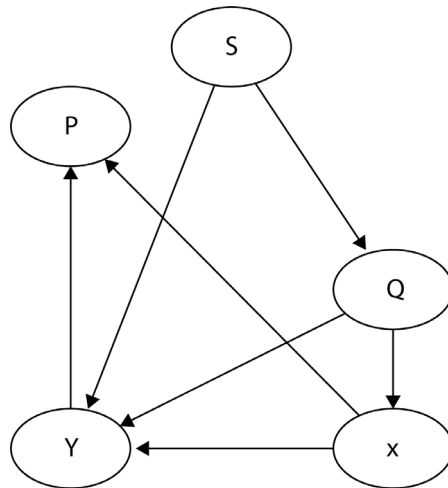


Figure 8.1 – The graphical model that we use in the experiment

Understanding the sample size – results

The results are presented in *Figure 8.2*. We see sample size on the x axis and percentage error on the y axis. 100% error means that the coefficient returned by the model was twice the size of the true coefficient; negative 100% means that the estimated effect was 0 (the true effect minus 100% of the true effect is 0):

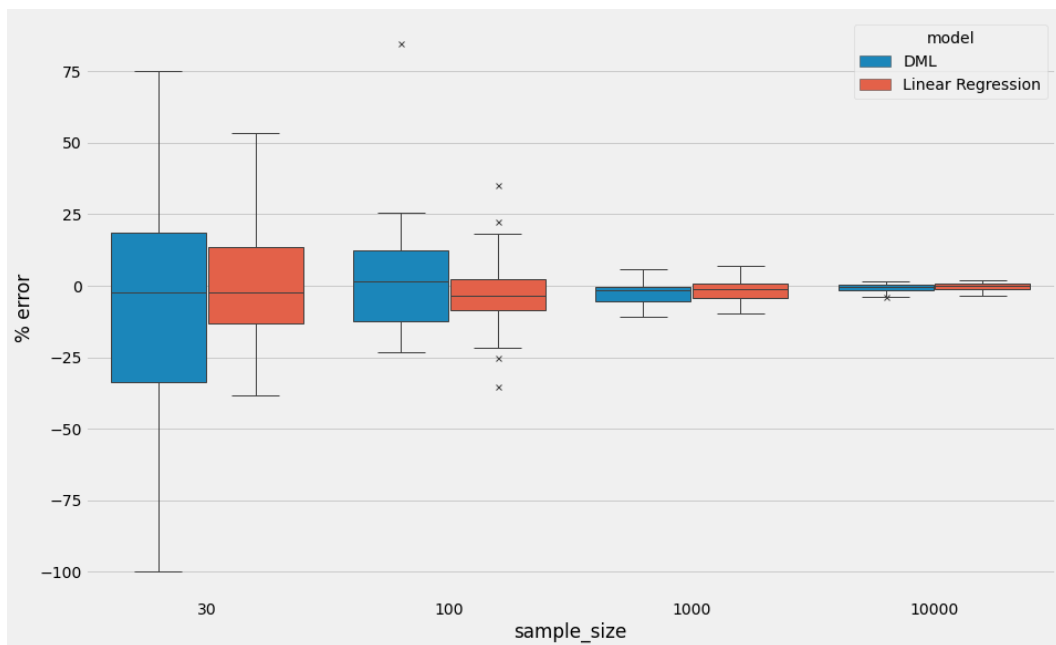


Figure 8.2 – The results of the sample size experiment

As we can see, for the smallest sample size, both models perform pretty poorly (despite the fact that the causal effect is identifiable in graphical terms). That said, the error variance of the DML model is significantly higher than the error variance of the simple linear model. At 100 observations, both models' performance match closely, and at 1,000 and 10,000 observations, the performance seems to converge. Note that we used a back-door criterion to identify estimands in this experiment. Back-door is computationally simple and much less challenging than front-door or other approaches.

To make the most out of smaller sample sizes, a good idea would be to bootstrap your results, if you can afford it.

Unverifiable assumptions

In some cases, it might be very difficult to find out whether you're meeting your assumptions. Let's start with confounding.

Imagine that you're trying to use the back-door criterion for your problem. In certain cases, you'll be able to rule out the possibility that there are other unobserved variables introducing confounding between your treatment and your outcome, but in many cases, it might be very difficult. In particular, when your research involves human interactions, financial markets, or other complex phenomena, making sure that the effects of interest are identifiable can be difficult if not impossible. That's likely one of the reasons for the huge popularity of **instrumental variable** techniques in contemporary applied econometrics. Although good instruments have a reputation of being notoriously hard to find, they might be easier to find than the certainty that we meet the back-door criterion.

An elephant in the room – hopeful or hopeless?

Some people reading about the challenges in causal inference from observational data might ask, "*Is there anything we can do?*"

I like the idea presented by James Altucher and Claudia Altucher in their book, *The Power of No*: if you cannot come up with 10 creative ideas, you should come up with 20 (Altucher & Altucher, 2014).

Let's see whether there's anything we can do to overcome these obstacles.

Let's eat the elephant

Let's talk about creative ideas for evaluating causal models.

The first level of creativity is to use the refutation tests that we described in the previous chapter. You already know how they work. One challenge with these tests is that they check for the overall correctness of the model structure, but they do not say much about how good the obtained estimate is.

By the way, if you have an idea for a new refutation test, go to <https://bit.ly/DoWhyRepo> and propose it in the discussion section or open a new pull request. You can help the causal community to move one step forward!

The second level of creativity is available when you have access to historical data coming from randomized experiments. You can compare your observational model with the experimental results and try to adjust your model accordingly.

The third level of creativity is to evaluate your modeling approach on simulated data with known outcomes. This might sound easy, but... if we had a reliable simulator of the process of interest, would we need machine learning in the first place?

Instead of building a simulator, we can try to learn it using generative neural networks. One such approach – **RealCause** – has been proposed by Brady Neal and colleagues (Neal et al., 2020). This approach certainly does not solve all your problems but can help you be more realistic, at least to an extent (for example, you can understand how your estimator behaves under **positivity assumption violation**).

An important fact about RealCause is that it generates realistic data distributions. It has been demonstrated that synthetic data can lead us astray when assessing the performance of causal models (Reisach et al., 2021; Curth et al., 2021). The good news is that the Python implementation of RealCause is open sourced and available at <https://bit.ly/RealCause>.

The fourth level of creativity is **sensitivity analysis**. In certain cases, we might have some idea about the magnitude of possible hidden confounding. In these cases, we can bound the error and check whether the estimated causal effect still holds if the confounders' influence reaches the maximum level that we think is reasonable to assume.

In other words, we can virtually check whether our effect would still hold in the worst-case scenario. Sensitivity analysis for regression models can be performed using the Python PySensemakr package (Cinelli and Hazlett, 2020; <https://bit.ly/PySensemakr>) or even using the online Sensemakr app (<https://bit.ly/SensemakrApp>).

Even better, the sensitivity analysis framework has been recently extended for a broad class of causal models (check Chernozhukov et al. (2022) for details).

In this section, we talked about the challenges that we can face when working with causal models and we discussed some creative ways to overcome them. We talked about the lack of causal graphs, insufficient sample sizes, and uncertainty regarding assumptions. We've seen that although there might not be a universal cure for all causal problems, we can definitely get creative and gain a more realistic understanding of our position.

In the next section, we'll discuss the positivity assumption.

Positivity

In this short section, we're going to learn about the **positivity assumption**, sometimes also called **overlap** or **common support**.

First, let's think about why this assumption is called *positivity*. It has to do with (strictly) **positive probabilities** – in other words, probabilities greater than zero.

What needs to have a probability greater than zero?

The answer to that is the probability of your treatment given all relevant control variables (the variables that are necessary to identify the effect – let's call them Z). Formally:

$$P(T = t | Z = z) > 0$$

The preceding formula must hold for all values of Z that are present in the population of interest (Hernán & Robins, 2020) and for all values of treatment T .

Let's imagine a simple example. In our dataset, we have 30 subjects described by one continuous feature Z . Each subject either received or did not receive a binary treatment T , and each subject has some continuous outcome Y . Additionally, let's assume that in order to identify the causal effect, we need to control for Z , which confounds the relationship between T and Y .

We can estimate the causal effect by computing:

$$E[Y|do(T = 1)] - E[Y|do(T = 0)]$$

Using the adjustment formula (Pearl et al., 2016), to compute these quantities from observational data, we need to control for Z .

Now imagine an extreme case where the support of Z (in other words, values of Z) does not overlap at all with the treatment values (hence the names *overlap* and *common support*; Neal, 2020). Figure 8.3 presents a graphical representation of such a case:

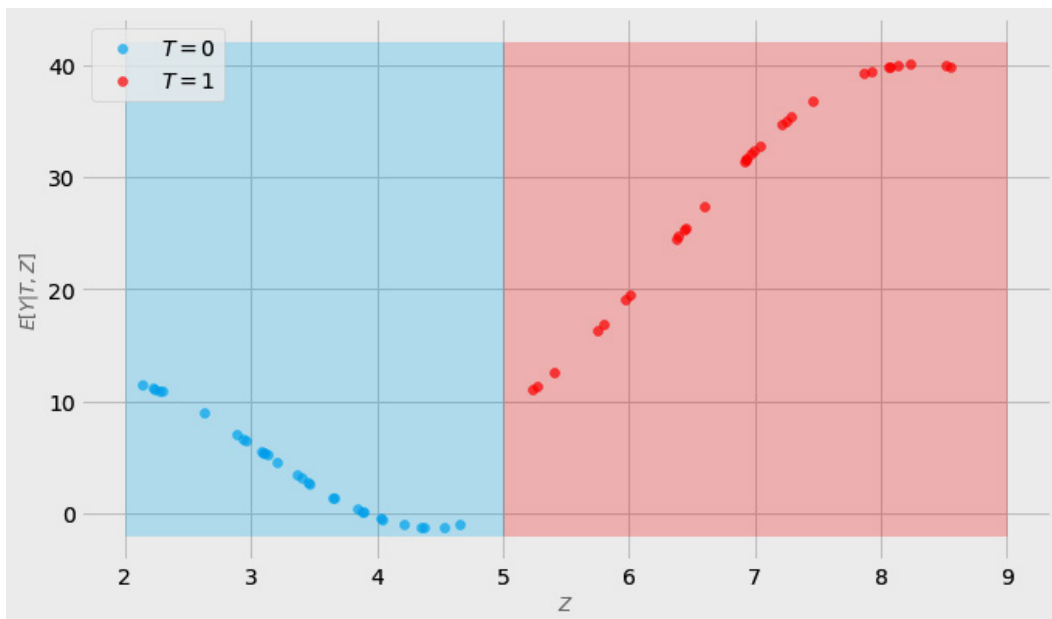


Figure 8.3 – Positivity violation example

In order to estimate the causal effect of the treatment T given Z , our estimator would need to extrapolate the red dots to the left (in the range between 2 and 5, where the support of Z for $T = 0$ is) and the blue dots to the right (in the range between 5 and 9, where the support of Z for $T = 1$ is). It's highly unlikely that any machine learning model would perform such an extrapolation realistically. *Figure 8.4* presents a possible extrapolation trajectory:

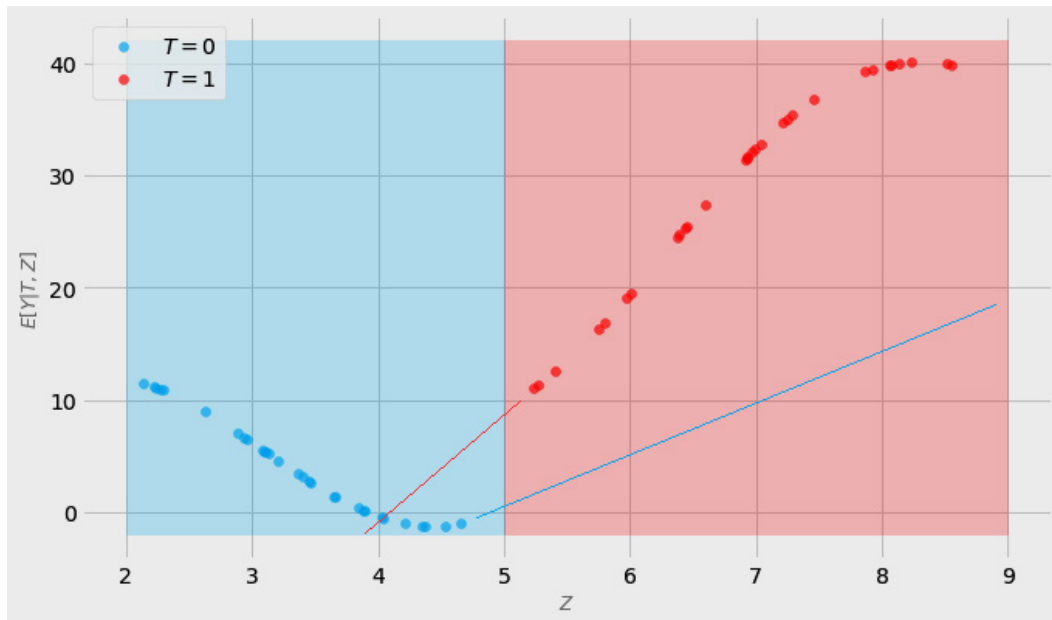


Figure 8.4 – Possible extrapolation trajectories (lines)

The red and blue lines in *Figure 8.4* represent possible extrapolation trajectories. How accurate do they seem to you?

To me, not very. They look like simple linear extrapolations, not capturing the non-linear qualities of the original functions.

Let's compare it to *Figure 8.5* where positivity is *met*:

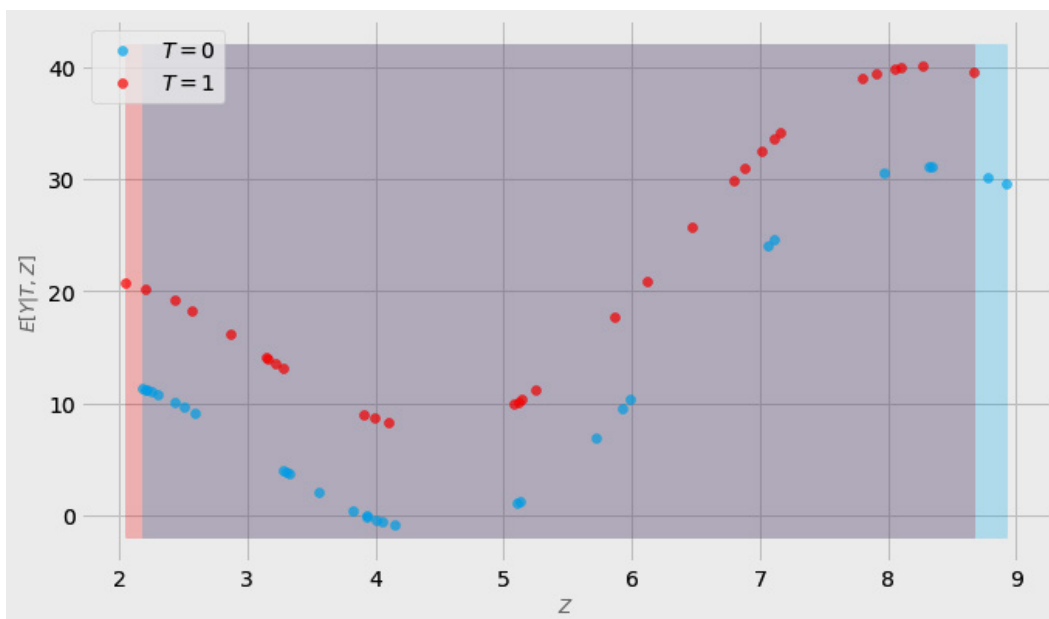


Figure 8.5 – Positivity assumption met

As you can see, here the model has a much easier task as it only needs to interpolate between the points.

Our example is very simple: we only have one confounder. In the real world, the data is often multidimensional, and making sure that $P(T = t|Z = z) > 0$ becomes more difficult. Think about a three-dimensional example. How about a 100-dimensional example?

As a side note, extrapolation and interpolation in high-dimensional spaces bring their own challenges (what does it mean to interpolate in an extremely high-dimensional space?). If this idea seems interesting to you, you might want to check the paper *Learning in High Dimension Always Amounts to Extrapolation* (Balestrieri et al., 2021) or listen to episode 86 of the *Machine Learning Street Talk* podcast with two of the authors of this paper – Randall Balestrieri and Yann LeCun (Machine Learning Street Talk, 2022).

In this section, we learned about the positivity assumption. We demonstrated two cases using a one-dimensional example: when the assumption is violated and when it holds. We also discussed how positivity is related to estimation and briefly noted the challenges that we can face in higher dimensions.

In the next section, we'll discuss the exchangeability assumption.

Exchangeability

In this section, we'll introduce the **exchangeability** assumption (also known as the **ignorability** assumption) and discuss its relation to confounding.

Exchangeable subjects

The main idea behind exchangeability is the following: the treated subjects, had they been untreated, would have experienced the same average outcome as the untreated did (being actually untreated) and vice versa (Hernán & Robins, 2020).

Formally speaking, exchangeability is usually defined as:

$$\{Y^0, Y^1\} \perp\!\!\!\perp T|Z$$

In the preceding formula, Y^0 and Y^1 are counterfactual outcomes under $T = 0$ and $T = 1$ respectively, and Z is a vector of control variables. If you're getting a feeling of confusion or even circularity when thinking about this definition, you're most likely not alone. According to Pearl (2009), many people see this definition as difficult to understand.

At the same time, the core idea behind it is simple: the treated and the untreated need to share all the relevant characteristics that can influence the outcome.

Can we express this in simpler terms?

Exchangeability versus confounding

Exchangeability is an idea that comes from the **potential outcomes** framework. Potential outcomes is a causal framework introduced by Donald Rubin in the 1970s (Rubin, 1974), but the core ideas go back to the 1920s and Polish statistician Jerzy Neyman's master thesis (Neyman, 1923).

In fact, the potential outcomes framework aims to achieve the same goals as SCM/*do*-calculus-based causal inference, just using different means (see Pearl, 2009, pp. 98-102, 243-245).

Pearl argues that both frameworks are logically equivalent and can be used interchangeably or symbiotically, pointing out that graphical models can help clearly address challenges that might be difficult to spot using the potential outcomes formalism (see <https://bit.ly/POvsSCM> for examples). I wholeheartedly agree with the latter.

We will not go deep into exchangeability/ignorability definitions here. Instead, we'll assume that – in general – *exchangeability* can be reduced to *unconfoundedness*. This means that we can achieve exchangeability using any of the deconfounding techniques we have introduced so far. This view is not always accepted, but further details are beyond the scope of this book.

For a more detailed discussion on the relationship between exchangeability and confounding, see Pearl (2009, pp. 196-199, 341-344) and Hernán & Robins (2020, pp. 27-31).

In this section, we discussed the exchangeability assumption. We introduced the potential outcomes definition of exchangeability and sketched its connection to the concept of confounding. In the next section, we're going to discuss three other assumptions that you can find in the causality literature.

...and more

In this short section, we'll introduce and briefly discuss three assumptions: the **modularity assumption**, **stable unit treatment value assumption (SUTVA)**, and the **consistency assumption**.

Modularity

Imagine that you're standing on the rooftop of a tall building and you're dropping two apples. Halfway down, there's a net that catches one of the apples.

The net performs an *intervention* for one of the apples, yet the second apple remains unaffected.

That's the essence of the **modularity assumption**, also known as the **independent mechanisms assumption**.

Speaking more formally, if we perform an intervention on a single variable X , the structural equation for this variable will be changed (for example, set to a constant), yet *all other* structural equations in our system of interest will remain *untouched*.

Modularity assumption is central to *do*-calculus as it's at the core of the logic of interventions.

Let's see an example.

Consider the following SCM $\mathcal{S}$:

$$X := \mathcal{N}(0, 1)$$

$$Z := \mathcal{N}(0, 1)$$

$$R := X + Z$$

$$Y := R + X$$

Figure 8.6 presents a graphical representation of $\mathcal{S}$.

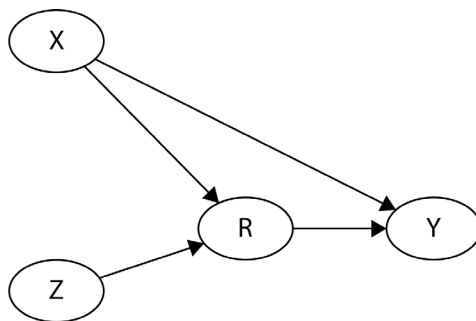


Figure 8.6 – The modularity example SCM $\mathcal{S}$

The modularity assumption says that if we intervene on node R by setting its value to r , the only equation that will change in $\mathcal{S}$ is the following:

$$R := r$$

All the other structural equations will remain the same.

The SCM $\mathcal{S}$ after the intervention, let's call it $\mathcal{S}_M$ (we often add a subscript M (modified) to denote an SCM after intervention), is presented in Figure 8.7:

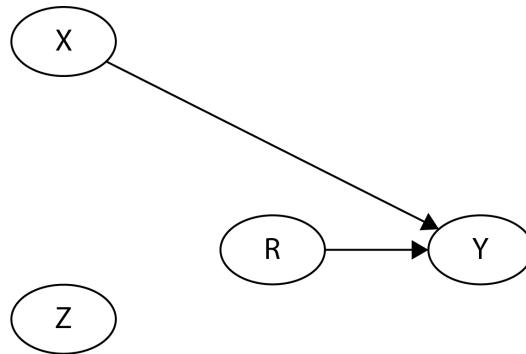


Figure 8.7 – The modified SCM $\mathcal{S}_M$

There are no longer edges from X and Z to R because we forced R to take value r and therefore any change in X or Z won't affect the value of R (this is known as a **perfect intervention**).

At the same time, the edge from X to Y is untouched (despite our intervention on R , changes in X will still affect the value of Y) in accordance with the modularity assumption.

Removing edges from the graph when we intervene on (a set of) variable(s) is sometimes called **graph mutilation**. I don't like this name too much. A more neutral term is **graph modification**.

From a graphical perspective, modularity can be summarized in the following way:

- When we perform a perfect intervention on node V , all V 's incoming edges are deleted in the modified graph
- No incoming edges to other (unintervened) nodes are modified

Another way to think about modularity is that changes caused by interventions are *local*: only the mechanism for the intervened variables change (we remove the incoming edges), but other mechanisms in the system remain unchanged.

Modularity might seem challenging to understand at first, but at its core, it's deeply intuitive.

Let's imagine a world where modularity does not hold.

In this world, you'd make an intervention on your web page by changing the button shape and as a result, your web page will also automatically be translated into Portuguese, while your computer will automatically start playing Barbra Streisand songs and your lawyer will immediately color their hair green – all without any other changes in the world besides your button shape. Such a world might be interesting but hard to navigate in terms of understanding causal mechanisms.

In this sense, modularity is a common sense assumption.

To summarize, the modularity assumption states that when we perform a (perfect) intervention on one variable in the system, the only *structural* change that takes place in this system is the removal of this variable's incoming edges (which is equivalent to the modification of its structural equation) and the rest of the system remains *structurally* unchanged.

SUTVA

The SUTVA is another assumption coming from the potential outcomes framework.

The assumption states that the fact that one unit (individual, subject, or object) receives treatment does not influence any other units.

This assumption might often be challenged in the context of interacting subjects. For instance, if one person in a household decides to start psychotherapy, other household members will likely also be affected (when one person changes their thinking and/or behavior, their environment usually reacts to this change in one way or another). Similarly, if you encourage some users of a social network to send more messages, the recipients of these messages are also likely to start sending more messages (not because they were encouraged by you, but rather because they want to respond to the senders).

At the experimental level, researchers are trying to overcome these challenges using various techniques such as cluster-level (rather than individual-level) randomization or so-called *ego-randomization* (Gui et al., 2015; Saint-Jacques et al., 2018). Some of these techniques can also be applied to observational data.

Consistency

The last assumption that we're going to discuss in this section is the **consistency** assumption. Consistency comes from the potential outcomes framework.

The assumption is also known as *no multiple versions of treatment* and is sometimes included as the second part of the SUTVA assumption (yes, I know, this might be confusing).

Let's get to the point.

Imagine that the treatment is to *win a car*. There are two levels of treatment: you either get a car or don't.

Some people in the treatment group win a brand-new electric BMW while others get a rusty, 20-year-old Mazda without wheels. If our outcome variable is the level of excitement, we'd expect that on average the same person's level of excitement would differ between the two versions of treatment. That would be an example of a violation of consistency as we essentially encode two variants of treatment as one.

Another way of looking at consistency comes from Pearl and Mackenzie. In their interpretation, consistency means that the experiment is "*free of placebo effect and other imperfections*" (Pearl & Mackenzie, 2019, p. 281).

If you only take two things out of our discussion on consistency, it should be these:

- Treatments should be well defined
- There should not be *hidden versions* of treatments

These are the two essential characteristics of consistency.

In this section, we discussed three assumptions: modularity (independent mechanisms), SUTVA, and consistency. The first one comes from the graphical/*do*-calculus framework, while the remaining two have their source in the potential outcomes framework.

In the next section, we'll shed some new light on the topic of unobserved variables.

Call me names – spurious relationships in the wild

Don't you feel that when we talk about spurious relationships and unobserved confounding, it's almost like we're talking about good old friends now? Maybe they are trouble sometimes, yet they just feel so familiar it's hard to imagine the future without them.

We will start this section with a reflection on naming conventions regarding bias/spurious relationships/confounding across the fields. In the second part of the section, we'll discuss selection bias as a special subtype of spuriousness that plays an important role in epidemiology.

Names, names, names

Oh boy! Reading about causality across domains can be a confusing experience! Some authors suggest using the term **confounding** only when there's a **common cause** of the treatment and the outcome (Peters et al., 2017, p. 172; Hernán & Robins, 2020, p. 103); others allow using this term also in other cases of **spuriousness** (Pearl & Mackenzie, 2019, p. 161).

Another term popular in the literature is **selection bias**. Definitions of selection bias differ between fields. Hernán & Robins (2020) advocate for a clear distinction between confounding (understood by them in terms of *common causes*) and selection bias (understood in terms of *common outcomes*).

Note that in the field of econometrics, the term *selection bias* might be used to describe any type of confounding bias.

Although it might be challenging sometimes, it's good to understand which convention a person you talk to (or read) uses.

Let's review what Hernán and Robins (2020) call a selection bias and learn two valuable lessons.

Should I ask you or someone who's not here?

It's 1943. American military planes are engaged in numerous missions and many of them are not coming back home. The ones that are coming back are often damaged.

An interesting fact is that the damage does not look random. It seems that there's a clear pattern to it. Bullet holes are concentrated around the fuel system and fuselage, but not so much around the engines.

The military engineers decide to consult a renowned group of statisticians called the **Statistical Research Group (SRG)** to help them figure out how to optimally distribute the armor so that the places that are at the highest risk of damage are protected.

A person who was faced with this question was Abraham Wald, a Hungarian-born Jewish mathematician, who worked at Columbia University and was a part of the SRG at the time.

Instead of providing the army with the answer, Wald asked a question.

It was a seemingly simple one: "*Where are the missing holes?*"

Missing holes?

What Wald meant were the holes that we've never observed. The ones that were in the planes that never came back.

DAG them!

What Wald pointed out is so-called **survivorship bias**. It's a type of selection bias where the effects are estimated only on a subpopulation of survivors with an intention to generalize them to the entire population. The problem is that the population might differ from the selected sub-sample. This bias is well known in epidemiology (think about disease survivors, for instance).

Let's put a hypothetical SCM together to represent the problem that Wald was facing.

Figure 8.8 is a graphical representation of this SCM:

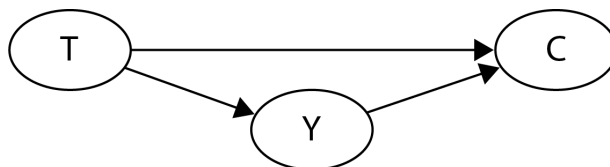


Figure 8.8 – An SCM representation of Wald's problem

We have three nodes: T , Y , and C . T represents the treatment, the number of enemy bullets shot at the engines, Y stands for the severity of plane damage, and C is a binary variable encoding whether a plane came back home (1) or not (0).

Note that what we do by only looking at planes that came back home is *implicitly conditioning on C* .

Two factors influence C : the number of bullets shot at the engines (T) and the overall damage severity (Y), which also depends on T . If the number of bullets shot at the engines is high enough, the plane is not coming back home regardless of other damage. It can also be the case that the number of bullets shot at engines is not that high, but the overall damage is so severe that the plane does not make it anyway.

Note that conditioning on C opens a spurious path, $T \rightarrow C \leftarrow Y$. This is precisely the source of bias because C is a *collider* on this path and – as you might remember – conditioning on a collider opens the information flow (as opposed to conditioning on chains and forks).

Let's translate our example into Python to see how this logic works on an actual dataset.

First, let's define the sample size and the set of structural assignments that defines our SCM:

```
SAMPLE_SIZE = 1000

# A hypothetical SCM
T = np.random.uniform(20, 110, SAMPLE_SIZE)
Y = T + np.random.uniform(0, 40, SAMPLE_SIZE)
C = (T + Y < 100).astype('int')
```

Let's put all the variables in a pandas DataFrame for easier manipulation:

```
df = pd.DataFrame(np.stack([T, Y, C]).T, columns=['T',
        'Y', 'C'])
```

Finally, let's condition on C and plot the results:

```
# Compare average damage (biased vs unbiased)
plt.figure(figsize=(8, 4))

plt.hist(df[df['C'] == 1]['Y'], label='Came back = 1',
        color=COLORS[0], alpha=.5)
plt.hist(df[df['C'] == 0]['Y'], label='Came back = 0',
        color=COLORS[1], alpha=.5, bins=25)

plt.xlabel('$Damage$ $severity$', alpha=.5, fontsize=12)
plt.ylabel('$Frequency$', alpha=.5, fontsize=12)

plt.legend()
plt.show()
```

Figure 8.9 presents the output of this block:

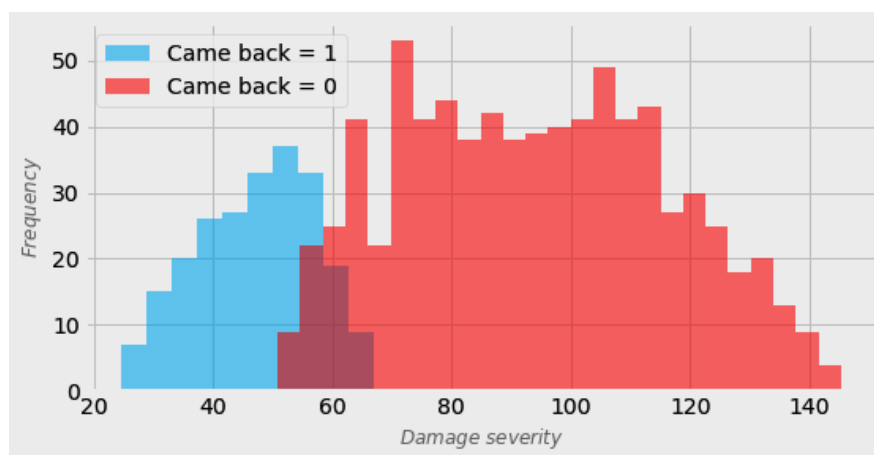


Figure 8.9 – A histogram of plane damage conditioned on C

The data in blue represents the damage severity for the planes that came back home. The data in red denotes the damage severity of the planes that did not come back.

As you can see, the severity is much higher for the planes that did not make it back home and there's only a small overlap between the two sets around the value of **60**.

Although our example is simplified (we don't take the damage location into account in our histogram), the conclusion is the same as Wald's – what matters the most are the *missing holes*, not the ones that we've observed. The first lesson is: look at what's missing.

If you want to understand the original solution proposed by Wald, you can find it in Wald (1980).

More selection bias

The DAG in Figure 8.8 represents just one possible combination of nodes leading to selection bias. Before we conclude this chapter, let's see some more examples. I picked something a bit spicier this time. Let's start with Figure 8.10:

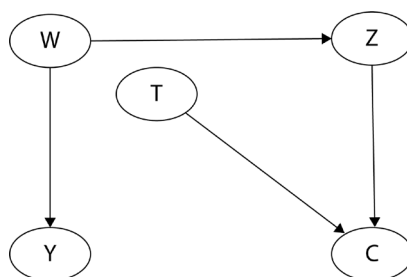


Figure 8.10 – An example of selection bias

The DAG in *Figure 8.10* is interesting. There is no connection between treatment T and the outcome Y , yet conditioning on C opens a path between the two. This is sometimes called **selection bias under the null** (Hernán and Robins, 2020).

Let's see one more example (*Figure 8.11*):

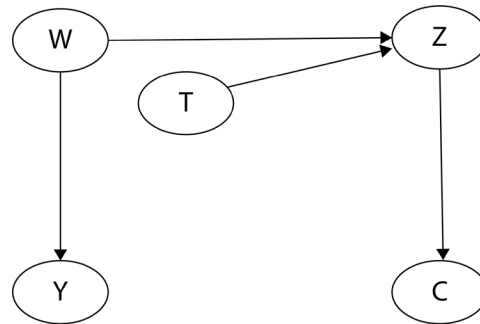


Figure 8.11 – An example of selection bias without the edge between T and C

The DAG in *Figure 8.11* is even more interesting. There are no edges directly connecting T and C or Y and C . Can you guess what's the source of spuriousness here?

Note that C is a child of Z . Controlling for C (by using it as a selection criterion) partially opens the path $T \rightarrow Z \leftarrow W \rightarrow Y$. We say *partially* because C usually *contains* some of Z 's variability.

This is a fact worth remembering – conditioning *on descendants* (yes, it's not only about children) of *colliders* usually *partially opens the path* on which the collider resides (in very special cases, conditioning on descendants could leave the path closed, but these cases are rather rare). The second lesson is look further than you think is necessary.

In this section, we discussed the concepts of selection bias, confounding, and spuriousness. We showed that naming conventions regarding these concepts differ across disciplines and subfields.

In the second part of the section, we focused on selection bias and recalled the story of Abraham Wald. We cast this story into the language of DAGs and translated it into Python to see how selection bias works in practice.

Two lessons we learned in this section are look at what's missing and look further than you think is necessary.

Wrapping it up

In this chapter, we talked about the challenges that we face while using causal inference methods in practice. We discussed important assumptions and proposed potential solutions to some of the discussed challenges. We got back to the topic of confounding and showed examples of selection bias.